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Course Code: CSE0001

Course Title: Fundamentals of AI and ML

1. Introduction

This project is about predicting student marks based on how many hours they study. Today, data is very important in making decisions, and this project shows how we can use simple machine learning to make predictions. The main aim of this project is to understand how basic AI/ML works in a simple and practical way.

2. Problem Statement

Many students do not know how much they should study to get good marks. Sometimes they study more but still don't get expected results. So, this project tries to give an idea of how study hours can affect marks.

3. Solution

To solve this problem, I used a simple machine learning method called Linear Regression. It helps to find a relationship between:

Study hours

Marks

Based on this, the model can predict marks when we enter study hours.

4. Approach

First, I created a small dataset with study hours and marks. Then I used Python to build the model.

Steps I followed:

Loaded the dataset

Selected study hours as input

Selected marks as output

Trained the model

Took input from user

Displayed predicted marks

5. Challenges Faced

At the beginning, it was difficult to understand how the model works. I also faced some confusion while setting up the code and libraries. But after practice, it became easier.

6. Learning Outcomes

From this project, I learned:

Basics of machine learning

How to use Python for simple projects

How prediction works

Importance of data